

AI-ML Intelligent Dead Reckoning System (IDR) - Complete Solution Overview

This document provides a complete narrative of the problem we are solving, our technical approach, and exactly what our team will deliver (the presentable solution) for ISRO Problem Statement 26168.

1. What Exactly is Our Solution Fixing?

The Core Problem:

Currently, navigation apps (like Google Maps) rely heavily on GPS/GNSS satellite signals. When a vehicle enters a tunnel, a dense forest, a deep urban canyon (between skyscrapers), or a multi-level parking lot, the satellite signal drops. When this happens, the navigation app stops updating accurately. It either freezes, jumps erratically, or miscalculates the vehicle's position.

The Challenge:

Modern luxury cars have built-in, factory-calibrated wheel sensors to keep tracking movement even without GPS. However, the vast majority of vehicles in India (trucks, older cars, two-wheelers) do not. They rely entirely on a smartphone mounted on the dashboard.

Our Fix:

We are building a system that allows a regular smartphone to continue tracking the vehicle's exact position **even when GPS completely fails**, using only the phone's internal motion sensors (accelerometer and gyroscope). We are fixing the "GPS blackout" problem for everyday vehicles without requiring expensive external hardware.

2. Our Solution Approach (How It Works)

We cannot just use the phone's raw motion sensors, because they are extremely noisy. Potholes, engine vibrations, and cheap sensor hardware cause exponential error (drift)—the phone might think it moved 100 meters when it only moved 10.

To solve this, our approach relies on **AI and Sensor Fusion**:

Step A: AI-Powered Speed & Vibration Filter

Instead of trying to calculate speed mathematically (which fails due to sensor noise), we use **Deep Learning (1D-CNN or LSTM)**.

- We train an AI model to look at the raw, vibrating sensor data and accurately predict the vehicle's actual forward speed.
- The AI acts as a smart filter, ignoring potholes, speed bumps, and engine idling.

Step B: In-Vehicle Alignment

A user might mount their phone crookedly. Our system will use Principal Component Analysis (PCA) and gravity vectors to automatically figure out which way the car is facing relative to the phone, so it knows which way is "forward."

Step C: Sensor Fusion (GNSS + INS)

We will use an advanced mathematical filter (like an Unscented Kalman Filter - UKF).

- **When GPS is available:** The filter uses GPS to correct and calibrate the phone's motion sensors constantly.
- **When GPS drops (Blackout):** The system instantly (in milliseconds) switches over to our AI model. The AI takes over, using the calibrated sensors to perform "Dead Reckoning" (calculating new positions based on the last known GPS location, heading, and AI-predicted speed).

Step D: Smart Map-Matching (Non-Holonomic Constraints)

Even with AI, minor drift happens. We constrain the vehicle to a digital map (OpenStreetMap). We apply a rule: *Cars cannot slide sideways or fly*. By mathematically snapping the calculated trajectory to the nearest logical road segment on the map, we eliminate the remaining drift.

3. The Presentable Solution (What the Team will Deliver)

For the final presentation and evaluation (SIH Finale), our team will deliver two main components:

Deliverable 1: The Mobile Application (The Primary Showcase)

A fully functional Android/iOS mobile application.

- **UI/UX:** It will look like a standard navigation app with a smooth, uninterrupted vehicle icon moving on a map.
- **Live Demonstration:** We will be able to mount a phone in a car, drive it into a GNSS-denied environment (like an underground parking lot or a simulated tunnel), and the app will continue to track the vehicle accurately with **less than 10% drift** (e.g., less than 5 meters of error over a 50-meter GPS blackout).
- **On-Device AI:** The AI model will run locally on the phone (using TFLite/ONNX) without needing an internet connection.

Deliverable 2: The Edge-Deployable Software Engine (The Core IP)

ISRO also wants to see that our algorithm isn't locked *only* to smartphones.

- **The Engine:** We will provide a standalone software engine (written in C++ or Python).
- **Capability:** This engine can be deployed on edge computers (like a Raspberry Pi or NVIDIA Jetson) installed in a vehicle, taking input from high-end, external IMU sensors (like FOG IMUs) running at 200Hz.
- **Proof:** This proves our underlying AI and Sensor Fusion algorithms are modular, industrial-grade, and ready for broader aerospace or terrestrial applications beyond just consumer smartphones.

4. Why Our Solution Wins

1. **No External Speedometer Needed:** It relies purely on the smartphone's IMU, drastically lowering the barrier to entry for the Indian market.
2. **AI-Driven Accuracy:** It replaces traditional, error-prone math with AI trained specifically on vehicle kinematics.
3. **Seamless Handoff:** The user experiences zero lag or stutter when entering a tunnel; the transition from GPS to AI Dead Reckoning is instantaneous.